

Distributional and Inflectional Properties of Quantifying Words in German: Abstract Agreement Chains Revisited*

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1. Introduction

Prenominal modifiers typically exhibit the following inflectional behavior where adjectives must have inflections but numerals cannot:

- (1) Adjectives:¹
 - a. Adj-INFL N
 - b. Det Adj-INFL N
- (2) Numerals:²
 - a. Num___ N
 - b. Det Num___ N

Here, I focus on certain non-numeral quantifiers, which have some intriguing distributional and inflectional properties: When in initial position, some have an inflection but others do not (3a-a'); when in non-initial position, that is, when they are preceded by a determiner, the ones with an inflection are grammatical but the ones without are ungrammatical (3b-b'):

- (3) Non-numeral quantifiers:
 - a. Q-INFL N
 - a'. Q___ N
 - b. Det Q-INFL N
 - b'. * Det Q___ N

As will become clear (see the footnotes below), this involves a complex empirical topic with a lot of details, including some variation. However, the absence of the pattern in (3b') holds in general – this is the focus of the current investigation.

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¹ Adjectives like *lila* 'purple' and *rosa* 'pink' are special in that they take optional inflections:

- (i) a. lila(-ne) / rosa(-ne) Kleider
purple-INFL / pink-INFL clothes
- b. die lila(-nen) / rosa(-nen) Kleider
the purple-INFL / pink-INFL clothes

Note, however, that these types of adjectives are not ordinary (native) adjectives. Furthermore, adjectives based on geographical names, so-called toponymic formations, like *Berliner* '(from) Berlin' as well as *prima* 'great', *sexy* 'sexy', and a few others are always uninflected (Duden 1995: 256-57).

² This does not hold for *ein* 'one' in general or *zwei* 'two' and *drei* 'three' in the genitive, where the inflection *-er* must be added to the latter two elements. This requirement is presumably a reflex of the Genitive Rule in German (see, e.g., Gallmann 1996).

2. Scope of the Study

In what follows, I discuss certain non-numeral quantifiers (4). These quantifying words presuppose pluralities, with mass nouns in the singular (4a) and count nouns in the plural (4b). These elements are in complementary distribution with each other:³

- (4) a. viel /wenig/genügend/sämtliches/etliches/einiges/mehreres/
much/little /enough /all /much /some /several/

genug /etwas/lauter /massig /haufenweise Obst
enough/some /nothing.but/in.masses/in.heaps fruit
- b. ganze/halbe/viele /wenige/genügende/sämtliche/etliche/einige/mehrere/
whole/half /many/few /enough /all /many /some /several/

genug /etwas/lauter /massig /haufenweise Äpfel
enough/ some /nothing.but/in.masses/in.heaps apples

As is clear from (4a) and particularly from (4b), some quantifiers are inflected but others are not; compare, for instance, *sämtlich-e(s)* ‘all’ vs. *genug* ‘enough’. This distinction will be of crucial importance below.

Before proceeding, note that these elements form constituents with their following noun. I illustrate this below with the Verb-Second effect and Preposition Phrases for two of the less well-known cases (all are googled examples):

- (5) a. und massig Arbeit ist das ja nun nicht.
and in.masses work is that PRT now not
- b. Sonstartet das wieder in massig Arbeit aus!
otherwise degenerates that again in in.masses work PRT
- c. Haufenweise Sachen sind ganz unten auf der Liste gelandet, wie Putzen.
in.heaps things are quite at.the.bottom on the list landed like cleaning
- d. Es gibt haufenweise Namen für haufenweise Sachen.
there are in.heaps names for in.heaps things

2.1. Complementary Distribution: Some Exceptions

There are some exceptions to this complementary distribution: First, some quantifiers can co-occur. For instance, there is the well-known case in (6a), where the singular variant is usually but not exclusively pronominal. The examples in (6b) involve *etwas* as a degree element meaning

³ This is not an exhaustive list of quantifiers in German. Below, I briefly discuss quantificational determiners like *jeder* ‘every’, but there are also some other declinable adjectival quantifiers similar to (4) (e.g., *zahlreich* ‘numerous’, *zahllos* ‘countless’, *unzählbar* ‘innumerable’, *gesamt* ‘entire’, *massenhaft* ‘in masses’), including some other indeclinable ones (e.g., *alle(r)hand* ‘rather many’). Note also that *ganz* ‘whole’ and *halb* ‘half’ in the singular are special, as they only combine with proper names in those instances (e.g., *ganz/halb Europa* ‘the whole/half of Europe’, see Roehrs 2026). For that reason, they are not listed in (4a). Finally, *mehreres Obst* in (4a) has a sortal reading, which could be rendered as ‘several kinds of fruit’.

‘somewhat’.⁴ Third, there are also some googled cases as in (6c-d), where the function of *massig* and *lauter* is not entirely clear but seems to be that of a degree word as well. Finally, *ganz* ‘whole’ and *halb* ‘half’ can, with certain qualifications, be preceded by other quantifiers (6e-f). The data below are organized in such a way that the left column shows the cases in the singular, and the right column presents the cases in the plural:

- | | | | | |
|-----|----|---|-----|--|
| (6) | a. | einiges wenig (Zubehör/Mobiliar)
some little equipment/furniture | a’. | einige wenige Gründe
some few reasons |
| | b. | etwas viel/wenig Obst
somewhat much/little fruit | b’. | etwas viele/wenige Äpfel
somewhat many/few apples |
| | c. | massig viel Geld/Geschmack/Erfolg
really much money/flavor/success | c’. | massig viele Videos/Anrufe
really many videos/calls |
| | d. | lauter viel Arbeit/Eis/Spaß
very? much work/ice/fun | d’. | lauter viele Dinge/Ideen/Gründe
very? many things/ideas/reasons |
| | e. | viele / etc. ganze/halbe Äpfel
many / etc. whole/half apples | | |
| | f. | eine ganze halbe Seite
a whole half page | | |

For the sake of completeness, I also point out that some of these quantificational elements can also occur with numerals, but there is a difference between specific numbers like *fünfzig* ‘fifty’ and vague numbers like *Hundert* ‘hundred’ (and *Dutzend* ‘dozen’, *Tausend* ‘thousand’), in cases where the latter are not preceded by *ein* ‘one’ or a different numeral:⁵

- | | | |
|-----|----|---|
| (7) | a. | *ganze/*halbe/*viele /*wenige/*genügende/?sämtliche/*etliche/*einige/mehrere/
whole/ half / many/ few / enough / all / many / some /several/ |
| | | *genug /*etwas/*lauter /*massig /*haufenweise fünfzig Studenten
enough/ some / nothing.but/ in.masses/ in.heaps fifty students |
| | b. | *ganze/*halbe/viele /wenige/*genügende/?sämtliche/etliche/einige/mehrere/
whole/ half /many/few / enough / all /many /some /several/ |
| | | *genug /*etwas/*lauter /*massig /*haufenweise Hundert Studenten
enough/ some / nothing.but/ in.masses/ in.heaps hundred students |

⁴ The degree word *viel* ‘much’ can also occur with *wenig(e)* ‘little/few’, provided these two elements are separated by *zu* ‘too’: *viel zu wenig(e)* ‘much too little/few’.

⁵ The examples with *ganze* in (7a) and (7b) are fine if *ganze* is interpreted as ‘just/merely’. Furthermore, *mehrere fünfzig Studenten* ‘several (sets of) fifty students’ in (7b) is fine but uncommon. Moreover, *etwa* ‘about’ (not to be confused with *etwas*) is fine before numerals. Note also that *lauter Hundert* in (7b) meaning ‘many hundreds’ sounds fairly good to me, but I have not been able to confirm this by examples identified through google.

Without discussion, I assume for the grammatical cases above that the first quantificational element is a modifier within a complex numeral, presumably similar to *fünf* ‘five’ in *fünfhundert* ‘five hundred’.

2.2. Differences Between Quantifiers and Quantifying Determiners

Before moving on, note that there is another set of quantifiers that can be distinguished from the ones just discussed above. These involve the quantificational determiners *ein* ‘a’, *kein* ‘no’, *jeder* ‘every’ (*jedweder* ‘every’, *jeglicher* ‘every’), *mancher* ‘some’, *alle* ‘all’, and *beide* ‘both’.⁶ There are two criteria that distinguish the non-numerical quantifiers discussed above from these quantificational determiners.

First, non-numeral quantifiers can be preceded by determiners (e.g., *das viele Obst* ‘(the) much fruit’), but quantificational determiners cannot (e.g., **der jede(r) Wagen* ‘(the) every car’).⁷ Given the complementary distribution in the latter case, quantificational determiners – as the name suggests – are usually analyzed as a type of determiner located in the DP-layer. By contrast, non-numeral quantifiers are often taken to be in CardP, a phrasal level below DP. Having said that, there is an important qualification here in that this criterion only applies to declinable non-numeral quantifiers – as documented in detail below, indeclinable non-numeral quantifiers cannot be preceded by determiners either (cf. (3b’) above). However, as I will argue in this paper, there is an independent explanation for that.

Second, quantificational determiners are followed by adjectives with weak inflections.⁸ As is well known, this is similar to determiners such as articles and demonstratives, which are

⁶ According to Milsark’s (1974) classification, *ein* ‘a’, *kein* ‘no’, and *mancher* ‘some’ are weak quantifiers (note, though, that many analyses treat *ein* ‘a’ as a semantically vacuous element, e.g., Coppock & Beaver 2015: 400-01, Heim & Kratzer 1998: 62, Kobele & Zimmermann 2012: 236). By contrast, all other quantificational determiners are strong quantifiers. As we will see below, strong quantifiers only take adjectives with weak inflections but weak quantifiers allow weak and/or strong inflections on the following adjectives. Having said that, there seems to be one potential exception as regards strong quantifiers. As stated by Kobele & Zimmermann (2012: 249-50), *all-* (in the singular) can exhibit parallel inflections with a following adjective: *all-er gut-er Rat* ‘all-ST good-ST advice’. They propose that *all-* is adjectival, located below the DP-level. Note that I find *all-* in other singular cases marked (e.g., *aller Zucker* ‘all sugar’). Here, I would prefer the alternative involving *ganz* ‘whole’ (e.g., *der ganze Zucker* ‘the entire sugar’). Thus, the case mentioned by Kobele & Zimmermann might be lexicalized in some way. Importantly, *all-* in the plural takes a weak adjective.

⁷ There is one exception involving *beide* ‘both’, which can do double duty as a quantificational determiner and a non-numeral quantifier: *beide* ‘both’ vs. *die beiden* ‘(the both =) both the’. Note that on the surface, *ein jeder* ‘(an every =) each’ appears to be another exception. However, *ein* here seems to function as an intensifying predeterminer (Roehrs 2025a: 57). This is presumably different from cases like *ein paar* ‘a couple’, *ein wenig* ‘a little’, and *ein bißchen* ‘a little bit’. Finally, note also that *manch* ‘some’ and *alle* ‘all’ are special in that they themselves can precede a determiner: *manch ein* ‘(some a =) some’, *alle diese* ‘all these’. None of these two-word combinations involving quantifiers will play a role in this paper. Consequently, they are not discussed here further.

⁸ As is well known, three instances of *ein* ‘a’ and *kein* ‘no’ – the forms in the nominative masculine and nominative/accusative neuter – are followed by strong adjectives. Observe now that there is an uninflected variant of inflected (*so*) *mancher* ‘some’. Unlike *ein* and *kein*, however, uninflected (*so*) *manch* ‘some’ is followed by a strong adjective more generally; for instance, in the plural, we find strings like *so manch klug-e Studenten* ‘some smart-ST students’. Note now that if this is also a (weak) quantificational determiner, then this one instance exclusively takes strong adjectives.

generally assumed to be in the DP-layer (since the facts of quantificational determiners followed by adjectives are well known, they are not illustrated here; see Roehrs 2025a and references cited therein for detailed discussion). With the exception of *ganz* ‘whole’ and *halb* ‘half’, which cannot be followed by an adjective (8a), non-numeral quantifiers always allow adjectives with strong inflections to follow. Note, though, that some inflected quantifiers also occur with weak adjectives (the cases involving variation are taken from Duden 2016: 364, 952; in (8b-c), I list the most common forms in the first data line; % indicates a googled example):⁹

- | | |
|---|--|
| (8) a. * <i>ganz</i> / <i>halb groß-e(-s)</i> Europa
whole / half big-WK(-ST) Europe | a'. <i>ganz-e</i> / <i>halb-e frisch-e</i> Äpfel
whole-ST / half-ST fresh-ST apples |
| b. <i>viel</i> / <i>wenig frisch-es</i> Obst
much / little fresh-ST fruit | b'. <i>viel-e</i> / <i>wenig-e frisch-e</i> Äpfel
many-ST / few-ST fresh-ST apples |
| b2. <i>viel-es brauchbar-e(-s)</i> Material
much-ST useful-WK(-ST) material | b2'. <i>viel</i> / <i>wenig frisch-e</i> Äpfel
many / few fresh-ST apples |
| c. <i>genügend frisch-es</i> Obst
enough fresh-ST fruit | c'. <i>genügend frisch-e</i> Äpfel
enough fresh-ST apples |
| c2. <i>genügend-es frisch-es</i> Obst
enough-ST fresh-ST fruit | c2'. <i>genügend-e frisch-e</i> Äpfel
enough-ST fresh-ST fruit |
| d. <i>sämtlich-es vorhanden-e(-s)</i> Material
all-ST available-WK(-ST) material | d'. <i>sämtlich-e wichtig-e(-n)</i> Meldungen
all-ST important-ST(-WK) messages |
| e. <i>etlich-es frisch-es</i> Obst
much-ST fresh-ST fruit | e'. <i>etlich-e frisch-e</i> Äpfel
many-ST fresh-ST apples |
| f. <i>einig-es brauchbar-e(-s)</i> Material
some-ST useful-WK(-ST) material | f'. <i>einig-e frisch-e</i> Äpfel
some-ST fresh-ST apples |
| g.% <i>noch mehrer-es frisch-es</i> Wasser
still several/more-ST fresh-ST water | g'. <i>mehrer-e frisch-e</i> Äpfel
several-ST fresh-ST apples |

Given the parallel inflections of the quantifiers and regular adjectives above, I take this as confirmation that these quantificational elements are indeed adjectival in nature. Consider the other set of quantifiers, which bear no inflections themselves and take adjectives with strong inflections only:

⁹ The instances of variation seem to be most common in the singular (neuter). Also, Kobele & Zimmermann (2012: 241) mark strings like *viel-er Zucker* ‘much-ST sugar’ and *viel-e Milch* ‘much-ST milk’ as ungrammatical (also Pafel 1994: 243). Furthermore, these two papers do not mention the possibility that *viel* can be uninflected in the plural. However, while perhaps less common, inflected forms of *viel* do occur in the singular and uninflected forms of *viel* do appear in the plural (see Duden 2016: 326-27, 952). Interestingly, unlike inflected *vieles* ‘much’ in (8b2), inflected *weniges* ‘little’ does not seem to allow a weak ending on a following adjective (see Duden 2016: 961-62). For plural *sämtliche* ‘all’, Duden (2016: 952) states that strong adjectives are actually rarer than weak ones. Since this inflectional variation of the following adjective is not crucial for the current paper, I have not investigated it in detail.

(9)	a.	genug frisch-es Obst enough fresh-ST fruit	a'.	genug frisch-e Äpfel enough fresh-ST apples
	b.	etwas frisch-es Obst some fresh-ST fruit	b'. ??	etwas frisch-e Äpfel some fresh-ST apples
	c.	lauter frisch-es Obst nothing.but fresh-ST fruit	c'.	lauter frisch-e Äpfel nothing.but fresh-ST apples
	d.	massig frisch-es Obst in.masses fresh-ST fruit	d'.	massig frisch-e Äpfel in.masses fresh-ST apples
	e.	haufenweise frisch-es Obst in.heaps fresh-ST fruit	e'.	haufenweise frisch-e Äpfel in.heaps fresh-ST apples

Quantificational determiners are fairly well known, and I will not discuss them here further. Rather, I focus on the non-numeral quantifiers that take strong adjectives, either the adjectival ones in (8) or the indeclinable ones in (9). Specifically, the question arises how to analyze the second set of quantifiers: Are indeclinable quantifiers lower in the structure like the adjectival elements in (8) or higher in the structure like quantificational determiners? This question will be tied to the explanation of why only declinable quantifiers can be preceded by determiners but indeclinable ones cannot. Before I turn to my proposal, I discuss the inflectional and distributional properties of these quantifiers in more detail.

3. Data

While there are a number of idiosyncratic properties of quantifiers (some of which presumably due to ongoing diachronic developments, see the various footnotes above), there is (at least) one generalization that can be established:

(10) Determiners cannot precede indeclinable (non-numeral) quantifiers.

To give substance to this generalization, we start with declinable adjectival quantifiers. Abstracting away from the special case of singular uninflected (11a), which does not tolerate adjectival modification and only allows quantifiable proper names (see Roehrs 2026), observe that the data in the (a)-type of examples may have an inflections, but the data in the (b)-type of examples must have an inflection (googled examples: (!) = possible without a following relative clause, where !, without brackets, indicates rarity; % = acceptable with a following relative clause; AR = archaic depending on the following noun where *Obst* ‘fruit’ in (17a) is fine in a sense of ‘several kinds of fruit’ but *Geld* ‘money’ in (17b) is archaic):¹⁰

¹⁰ Note that unlike *die ganzen Äpfel* ‘all apples/the whole[uncut] apples’ in (11b’), *ganze Äpfel* in (11a’) cannot mean ‘all apples’, but it can mean ‘whole[uncut] apples’ if *ganz* is stressed. Furthermore, *viel* ‘much’ in (12) has a suppletive form in the superlative (*meist-* ‘most’). Kobele & Zimmermann (2012: 257) state that due to this morphological form, the definite article must be present in this instance.

- | | | | | |
|------|--------------------|---|---------|--|
| (11) | a. | ganz*(-es) / halb*(-es) Europa | a'. | ganz*(-e) / halb*(-e) Äpfel |
| | b. | das ganz*(-e) / halb*(-e) Europa | b'. | die ganz*(-en) / halb*(-en) Äpfel |
| (12) | a. | viel([?] -es) / wenig([?] -es) Obst | a'. | viel(-e) / wenig(-e) Äpfel |
| | b. | das viel*(-e) / wenig*(-e) Obst | b'. | die viel*(-en) / wenig*(-en) Äpfel |
| (13) | a. | genügend([?] -es) Obst | a'. | genügend([?] -e) Äpfel |
| | b. (!) | die mehr als genügende Ausbeute | b'. (!) | die mehr als genügenden Bestände |
| (14) | a. | sämtlich*(-es) Obst | a'. | sämtlich*(-e) Äpfel |
| | b. | das sämtlich*(-e) Obst | b'. | die sämtlich*(-en) Äpfel |
| (15) | a. | etlich*(-es) Obst | a'. | etlich*(-e) Äpfel |
| | b. ! | das etlich*(-e) Leid | b'. | all die etlich*(-en) Jahre |
| (16) | a. | einig*(-es) Obst | a'. | einig*(-e) Äpfel |
| | b. ! | das einiger*(-e) Obst | b'. % | die einiger*(-en) Autos die 10 Mio kosten |
| (17) | a. ^(AR) | mehrer*(-es) Obst | a'. | mehrer*(-e) Äpfel |
| | b. ^{AR} | das mehrer*(-e) Geld | b'. % | die mehrer*(-en) Leute die seekrank wurden |

Kobele & Zimmermann (2012: 227) state that many – what they call – D-quantifiers can co-occur with a (primarily definite) determiner (e.g., *viel* ‘much’). There are two points to be made here: First, this means that D can presumably not stand for the syntactic category of determiner in this context. Second, this statement suggests that indefinite articles are not impossible, which is true (e.g., *ein genügendes Sprachverständnis* ‘(a) sufficient language comprehension’).

More importantly, with the exception of *ganz* and *halb* in the singular, all declinable quantifiers exhibit inflections in determinerless contexts, at least in some environments. In other words, there are no quantifiers that have no inflection in contexts where the determiner is missing, but inflections in contexts where the determiner is present. Thus, the distributional combination of Q₁___ N & Det Q₁-INFL N is not found. Note that the reverse (i.e., Q₁-INFL N & Det Q₁___ N) is not possible either. The latter, where an inflection is missing on a quantifier preceded by a determiner, is ruled out by the explanation offered below.

There are a number of details worth highlighting again. First, *ganz* ‘whole’ and *halb* ‘half’ cannot have an ending in the singular (11a) but must have an ending if preceded by a determiner (11b). Note that these are the only two (declinable) elements that cannot have an inflection in the singular (again, see Roehrs 2026). Second, *viel* ‘much/many’ and *wenig* ‘little/few’ (and *genügend* ‘sufficient’) have, with some qualification, optional endings but must have an ending if preceded by a determiner.¹¹ Like *ganz* and *halb*, these elements have some special inflectional properties. Unlike *ganz* and *halb*, they are the only (declinable) elements that can occur without an ending in the plural. Note, though, that Duden (2016: 327) points out that there is a difference in meaning with plural *viel(e)* ‘many’ and *wenig(e)* ‘few’ such that the inflected form is

¹¹ As documented above, *viel(e)* and *wenig(e)* have very similar (albeit not identical) properties. The distinctions presumably stem from their different histories (e.g., Roehrs & Sapp 2016).

distributive (“vereinzelt”) and the uninflected form is collective (“zusammenfassend”) in interpretation. These two readings are not possible with (singular) mass nouns, which are not countable and thus cannot have a distributive interpretation. Here, the inflectional variation seems to be due to individual preferences.¹²

To sum up, initial quantifiers either have optional inflections (e.g., *viel*-) or obligatory inflections (e.g., *sämtlich*-) in singular and plural contexts. However, when a determiner is present, the optional inflections become obligatory.

Like numerals, I assume that these quantifiers are in Spec,CardP (see Julien 2005), and that determiners, if present, are in DP:

(18) [D [[QP] Card [Agr [Num [N]]]]]

By contrast, *genug* ‘enough’, *etwas* ‘some’, and *lauter* ‘much/many’ are indeclinable and cannot take inflections at all (including in the genitive). Consider the (a)-type of examples below. Crucially, these quantifiers cannot be preceded by a determiner, as can be seen in the (b)-type of examples:¹³

- | | | | | |
|------|------|---------------------|-----|-------------------------|
| (19) | a. | genug Obst | a'. | genug Äpfel |
| | b. * | das genug(-e) Obst | b'. | * die genug(-en) Äpfel |
| (20) | a. | etwas Obst | a'. | ?? etwas Äpfel |
| | b. * | das etwas(-e) Obst | b'. | * die etwas(-en) Äpfel |
| (21) | a. | lauter Obst | a'. | lauter Äpfel |
| | b. * | das lauter(-e) Obst | b'. | * die lauter(-en) Äpfel |

¹² As pointed out by Duden (2016: 326), the Genitive Rule also applies to *viel* ‘much/many’ and *wenig* ‘little/few’. In other words, in genitive contexts, the relevant endings become obligatory. Importantly, no ending is possible on *ganz* ‘whole’ and *halb* ‘half’ in the singular or on other more generally indeclinable elements (see below).

¹³ Some of these elements have special properties; for instance, *genug* ‘enough’ can not only precede but also follow nouns (ia). However, this element must follow adjectives (ib). In order to avoid violations of the Head-Final Filter (Williams 1982), it can, for some speakers, take on an inflection when preceding a noun (ic) (Haider 2022; for similar facts in Dutch, see van Riemsdijk 1998: 673-74). It is noteworthy that (id) does not sound fully ungrammatical (cf. (19b) in the main text). In fact, one speaker I consulted found *das genuge Obst* in (19b) above to be quite good (but confirmed that uninflected *genug* preceded by a determiner is impossible). These facts may be related to the existence of (ic), where *genug* seems to exhibit features of an adjective:

- (i)
- | | | | |
|-------|---------------|-------------|-------|
| a. | Geld | genug | |
| | money | enough | |
| b. | groß | genug | |
| | big | enough | |
| c. % | groß | genug-e | Autos |
| | big | enough-INFL | cars |
| d. *? | das mehr als | genug-e | Obst |
| | the more than | enough-INFL | fruit |

Furthermore, similar to *genug*, *etwas* can also be a degree element as in (ib) here meaning ‘somewhat’, but it must precede its modifiee (see (6b) above). Interestingly, if the interpretation of the data in (6c-d) above is on the right track, then *lauter* ‘nothing but’ and *massig* ‘in masses’ seem to be able to function as degree elements as well.

(22) a. massig Obst a'. massig Äpfel
 b. * das massig Obst b'. * die massig Äpfel

(23) a. * fünfzig massig / massig fünfzig Äpfel
fifty in.masses / in.masses fifty apples
b. (die) fünfzig massig-e(-n) Äpfel
the fifty massive-ST(-WK) apples

(24) a. haufenweise Obst a'. haufenweise Äpfel
b. * das haufenweise Obst b'. * die haufenweise(-n) Äpfel

To recap this section, inflecting quantifiers may have an inflection, at least optionally, when in initial position (except for *ganz* ‘whole’ and *halb* ‘half’ in the singular) but must have an inflection when preceded by a determiner. By contrast, indeclinable quantifiers are fine in initial position but ungrammatical when preceded by a determiner. There are three questions to be answered now: (i) What makes the optional endings on the quantifiers obligatory when a determiner precedes, (ii) what explains the ungrammaticality of indeclinable quantifiers when in non-initial position, and (iii) what is the structural position of indeclinable quantifiers? In order to answer these questions, I take up the generalization provided above in more detail in the next sections.

The distributional and inflectional behavior documented in the last section can be represented more schematically in (25), repeated from the introduction and presented here in a rearranged order:¹⁴

9

(25) Non-numeral quantifiers:

- a. Det Q-INFL N
- b. * Det Q___ N
- c. Q-INFL N
- d. Q___ N

Returning to the generalization for the pattern in (25) from the previous section in (26a) below, note that it can also be given in its passivized (truth-conditionally equivalent) form in (26b), where the focus is more on the quantifier:

- (26) a. Determiners cannot precede indeclinable quantifiers.
- b. Indeclinable quantifiers cannot be preceded by determiners.

Note that the generalization in (26b) above can be restated as a material implication: If there is an indeclinable quantifier, then the latter cannot occur with a determiner (27a). Leaving out the two negatives in (26a) above, we can state alternatively that if there is a determiner, then the latter can only occur with a declinable quantifier (27b):

- (27) Provided Q is present:
- a. indeclinable Q $\rightarrow$ no determiner
- b. determiner $\rightarrow$ declinable Q

Both generalizations are logically equivalent, as can be seen in the relevant truth tables for material implication. Starting with the simpler case in (27b) above, consider the truth table in (28a) and compare that to the truth table for (27a) above in (28b):

(28) a. $p \rightarrow q$ (= (27b))			b. $\neg q \rightarrow \neg p$ (= (27a))				
Presence of Det: p Declinable Q: q			Absence of Det: $\neg p$ Indeclinable Q: $\neg q$				
p	q	$p \rightarrow q$	p	q	$\neg p$	$\neg q$	$\neg q \rightarrow \neg p$
T	T	T	T	T	F	F	T
T	F	F	T	F	F	T	F
F	T	T	F	T	T	F	T
F	F	T	F	F	T	T	T

Note that (28b) is a contrapositive of (28a). In other words, these are different formulations of the same basic generalization. I turn to the explanation of this generalization.

5. Explanation

Why can determiners not precede indeclinable quantifiers like *genug* ‘enough’, *etwas* ‘some’, *lauter* ‘many’, and others? I start with a previous proposal, showcasing some shortcomings as regards the facts above. Then I entertain two possible hypotheses to explain the facts: (I) indeclinable quantifiers are bimorphemic, with one part in D, thus blocking the presence of

determiners; (II) indeclinable quantifiers disrupt an agreement chain between determiners and head nouns. Given certain points, the second hypothesis seems to be preferable.

5.1. Previous Proposal

Pafel (1994: 245-46) documents certain semantic differences between initial *viel* (i.e., cases where a determiner is absent) and non-initial *viel*, for instance, with regard to the scope of negation. He proposes the structure in (29a) for initial *viel* and the structure in (30b) for non-initial *viel*. Note that e_D in (29a) stands for an empty determiner, which has the quantificational adjective right-adjoined to it. This adjunction structure forms a complex lexical entry of the category D:

- (29) a. [DP [D e_D *viele_A*] [NP *Äpfel*]]
 b. [DP *die* [NP *vielen* [NP *Äpfel*]]]

The inflectional variation documented above for initial *viel*, both in the singular and plural, may follow from the lexical account in (29a). Furthermore, if *viele* in (29b) is indeed an adjective, the inflection on it is also expected to be obligatory.¹⁵ Notice, however, that Pafel does not discuss *genug* ‘enough’, *lauter* ‘nothing but’, *massig* ‘in masses’ or *haufenweise* ‘in heaps’, and only briefly mentions *etwas* ‘some’ (and *was* as a pronominal element meaning ‘something’) (see his pages 270-71). Having said that, these elements could be analyzed as in (29a) above. Note in this regard that uninflected elements like initial *viel* are analyzed in this way. However, the question arises why indeclinable quantifiers cannot be analyzed as in (29b) above? Specifically, if these elements are generally categorized as adjectives in (29a), then (29b) should, at least in principle, be possible as well. If so, the ungrammaticality of these indeclinable quantifiers when preceded by a determiner becomes mysterious.

Before I turn to my own two hypotheses, note that the two types of quantifiers do not seem to involve a clear semantic difference, where the left column in (30) shows the declinable quantifiers and the right column provides the indeclinable counterparts:

- (30) a. genügend genug
 sufficient enough
 b. einige etwas
 some some
 c. viel(e) lauter, massig, haufenweise
 much/many nothing.but in.masses in.heaps

¹⁵ Pafel (1994: 259-61) accounts for the strong/weak alternation of adjective inflections by the feature [$\pm$ DEF], which might be syntactic in nature (his page 272). Specifically, determiner elements that are [+DEF] take a weak adjective, and elements that are [-DEF] take a strong adjective. Note, however, that (indefinite) *ein* ‘a’ and *kein* ‘no’, and (definite) possessive determiners like *mein* ‘my’ are treated as idiosyncratic exceptions. Furthermore, as shown in detail in Roehrs (2025a), a simple opposition of indefinite vs. definite cannot account for all cases (e.g., *ich Arm-er* ‘(I =) poor-ST me’ vs. *wir Arm-en* ‘(we =) poor-WK us’).

Considering the meanings, these elements seem to involve (near-)synonyms, which makes a semantic account to explain the above differences unlikely.

5.2. Hypothesis I: Bimorphemic Structure

All indeclinable quantifiers appear to be bimorphemic: *ge-nug*, *et-was*, *laut-er*, *mass-ig*, *haufen-weise*. It could be claimed then that one of these two parts is located in D, thus blocking the presence of a determiner. Illustrating with *etwas*, it is clear that *et-* is not in D (e.g., *die et-lichen* ‘(the) many’, *die et-waigen* ‘the possible’). Rather, *et-* is quantificational and can be assumed to be in Card (note that *ge-* is also quantificational in the sense of ‘collective’). Now, if *et-* is quantificational, then *was* is not in Card (the latter is already occupied by *et-*), it is not in N (cf. *welches/wessen Buch* ‘which/whose book’), but rather *was* is in D. Notice in this regard that Roehrs & Sapp (2016: 238-39) argue for this to hold in Old High German and Middle High German such that *was* in D undergoes Lowering onto *et*, bringing about the surface form *etwas*.¹⁶

It is noteworthy that this explanation would allow numerals and adjectives to occur with determiners, as the former two elements presumably do not have a morpheme in D.

Issue #1:

Involving two morphemes can only be a necessary (but not sufficient) condition, as other bimorphemic quantifiers do allow determiners (e.g., *die et-lich-en Leute* ‘(the) many people’, *die wen-ig-en Leute* ‘(the) few people’). However, we could claim that *et-* and *wen-* are quantificational and that *-lich* and *-ig* are adjective-forming elements (note in this regard that the latter type of quantifiers bears an adjectival inflection). This would mean that no morpheme is in D with these inflected adjectival elements, allowing determiners to occur. Consequently, while not a sufficient condition, this might still provide an explanation for the indeclinable cases.

Issue #2:

Having said that, note that some apparent morphemes do not exist independently (e.g., *-nug*) and that the analysis of other morphemes (e.g., *laut*, *-er*) is, at best, not obvious and, at worst, rather dubious.¹⁷ This undermines postulating a general bimorphemic structure.

Issue #3:

Also, the bimorphemic structure above may not hold for Modern German to begin with, as indefinite pronouns tend to undergo grammaticalization. For instance, Roehrs & Sapp (2016) propose for Modern German that *etwas* is in CardP. Observe that the latter claim will not explain the absence of the determiner with these quantifiers – there is no (structural) complementary distribution between the two relevant elements. Alternatively, we could suggest that the

¹⁶ By contrast, Roehrs (2008: 23) and references cited therein assume that *et-* is in D and that *was* is lower in the structure.

¹⁷ Notice that words with a stem related to *genug* ‘enough’ show the unlauted form of *-nug* (e.g., *zur Genüge* ‘enough’, *genügen* ‘be enough’), but each also includes *ge-*. Also, *lauter* ‘nothing but’ derives historically from the monomorphemic adjective *lauter* meaning ‘pure’ (Eckhardt 2006).

indeclinable quantifiers are in the DP-level, and similar to *so manch* ‘some’ (see Footnote 8), they would take strong adjectives. Note, though, that inflected quantifiers can be in either DP (e.g., *jed-er* ‘every’) or CardP (e.g., *viel-e* ‘many’). The question arises why indeclinable quantifiers must be in DP but cannot be in CardP. The postulation of the sole option of DP for the latter cases seems to involve a stipulation to account for the complementary distribution with determiners and does not present a real explanation.

Issue #4:

Finally, this type of analysis cannot explain why monomorphemic quantifiers like *viel* ‘much/many’ must take an inflection when preceded by a determiner.

5.3. Hypothesis II: Agreement Chain

Bayer (2015) proposes for a different set of data in German that non-agreeing elements like Saxon Genitives *disrupt* an agreement chain established between quantificational predeterminers and head nouns:¹⁸

- (31)
- | | | | |
|------|---------------|---------------|---------------|
| a. | <i>seine</i> | <i>Arbeit</i> | |
| | +AGR | +AGR | |
| b. | <i>alle</i> | <i>seine</i> | <i>Arbeit</i> |
| | +AGR | +AGR | +AGR |
| c. | <i>Peters</i> | <i>Arbeit</i> | |
| | –AGR | +AGR | |
| d. * | <i>alle</i> | <i>Peters</i> | <i>Arbeit</i> |
| | +AGR | –AGR | +AGR |

There are potential issues with the account of the data above utilizing an agreement chain that involves overt inflections.¹⁹ However, the data under investigation here seem to manifest the disruption of a chain lower in the structure (32a-b):

- (32)
- | | | |
|------|--------------|----------------------------|
| a. | <i>genug</i> | <i>Arbeit</i> |
| | –AGR | +AGR |
| b. * | <i>die</i> | <i>genug</i> <i>Arbeit</i> |
| | +AGR | –AGR +AGR |

This type of explanation readily extends to cases like *viel*, where the presence of a determiner makes the inflection obligatory. Compare (33a) to (33b-c):

¹⁸ English is different: *all Peter’s friends* is fully grammatical.

¹⁹ Bayer’s (2015) proposal faces challenges with uninflected predeterminer elements like *all* ‘all’, *dies* ‘this’, *von Peter* ‘of Peter’, and possibly also with uninflected instances of determiner elements like *sein* ‘his’, etc. In addition, genitives do not disrupt the chain if they occur lower in the structure as in Old High German or Early New High German. In other words, besides “overgeneration”, where Bayer’s chain does not rule out some relevant ungrammatical cases, there are also cases of “undergeneration”, where it rules out grammatical cases (for detailed discussion, see Roehrs 2025b).

- (33) a. viel / viele Arbeit
 –AGR / +AGR +AGR
 b. * die viel Arbeit
 +AGR –AGR +AGR
 c. die viele Arbeit
 +AGR +AGR +AGR

To be clear, this type of analysis explains both the absence of determiners with indeclinable quantifiers and the obligatory presence of (optional) inflections on declinable quantifiers when a determiner is present.

The quantifiers discussed here all seem to be weak in the sense of Milsark's (1974) classification. This is different from quantificational determiners, which are strong (except for *ein* 'a', *kein* 'no', and *manch* 'some'). Given this and the current analysis, we can suggest that indeclinable quantifiers are also located in Spec,CardP. This has an interesting consequence.

Eckardt (2006: 204) claims that *lauter* 'nothing but' is a determiner derived from an old adjective meaning 'pure'. Syntactically, Eckhardt argues that it is a determiner(-like) word, but semantically, she suggests that *lauter* has remained an adjective (her page 233). With the discussion above in mind, we can suggest that *lauter* is not a determiner, but rather has remained adjectival in the syntax as well. The complementary distribution with determiners has an independent explanation.

The agreement chain does not apply to all elements: There is no disruption with *wie* 'how' or *so* 'so'. These lexical items are degree words modifying the quantifiers in Spec,CardP. As such, they are not relevant for the workings of the chain (indicated below by putting AGR in brackets):²⁰

- (34) a. wie viel Arbeit
 [–AGR] –AGR +AGR
 b. so viel Arbeit
 [–AGR] –AGR +AGR

Issue #1:

Like above, there is no disruption with *Peters* in (35a-b). However, the question arises why (35a) is worse than (35b):

- (35) a. ?? Peters viel Arbeit
 –AGR –AGR +AGR
 b. (?) Peters viele Arbeit
 –AGR +AGR +AGR

²⁰ Modifying elements are often taken to be in the specifier position of their modifiee; that is, they are in a different structural position. Note in this regard that sensitivity to structure is independently needed for Bayer's agreement chain: As discussed by Bayer (2015), cases like *alle der Maria ihre Bücher* 'all (the Maria her =) Maria's books', where the dative phrase *der Maria* is assumed to be in Spec,DP, are fine.

The Saxon Genitive *Peters*, specifically the possessive morpheme *-s*, is not an agreeing element; rather, *Peters* is a (preceding) determiner-like element. Observe that this is an interesting data point, as it distinguishes between “non-agreeing” determiner elements (*Peters*) and “agreeing” determiner elements (*die* ‘the’, *seine* ‘his’, etc.).

Proposal: Above, I illustrated that –AGR can be on the left in a two-member chain (e.g., *Peters Arbeit* ‘Peter’s work’). As is evident from (35b), this must also be possible in a three-member chain. However, it seems clear that the chain cannot have two adjacent –AGR on the left (35a). This is different from disrupting the agreement chain, but it might be a necessary extension. In fact, if this chain is about overt inflections, then this extension is also needed to explain the following contrast:

- (36) a. * mein viel Sport
 –AGR –AGR +AGR
 b. mein vieler Sport
 –AGR +AGR +AGR

Issue #2:

This account seems to rule out numerals (e.g., *die zwei Autos* ‘the two cars’) and special types of adjectives (e.g., *das lila Auto* ‘the purple car’) when preceded by a determiner.

Proposal: The chain seems to be restricted to certain lexical categories. If Roehrs (2025b) is right in his critique of Bayer (2015), then this restriction does not apply to Saxon Genitives, which involve open-class elements, but rather it concerns non-numeral quantifiers, which present closed-class lexical items.

Issue #3:

I return to the discussion of the distribution of the quantifier *haufenweise* ‘in heaps’ from above:

- (37) a. haufenweise Obst a’. haufenweise Äpfel
 b. * das haufenweise Obst b’. * die haufenweise(-n) Äpfel

Recall that I mentioned above that this element has some special properties.

Pittner (1995: 36-37) provides data similar to (37b) above as in (38a) below. Crucially, however, she notes that nominalized verbs as *Horten* ‘hoarding’ in (38b) do allow this (and similar) quantifiers to intervene when a determiner is present. This yields an intriguing contrast between (38a) and (38b). Also, if an adjective is inserted between this quantifier and a regular noun, the example in (38a) becomes grammatical. Compare (38a) to (38c):

- (38) a. * die haufenweise/massenweise/stapelweise/eimerweise/kübelweise/kistenweise/
 the in.heaps /in.masses /in.piles /in.buckets /in.pails /in.bboxes
 bergeweise Bücher
 in.mountains books

- b. das haufenweise/massenweise/stapelweise/?eimerweise/?kübelweise/kistenweise/
 the in.heaps /in.masses /in.piles /in.buckets /in.pails /in.boxes
 bergeweise Horten von überflüssigen Büchern
 in.mountains hoarding of spare books
- c. die haufenweise/massenweise/stapelweise/?eimerweise/?kübelweise/kistenweise/
 the in.heaps /in.masses /in.piles /in.buckets /in.pails /in.boxes
 bergeweise überflüssigen Bücher
 in.mountains spare books

Pittner analyzes *haufenweise* as a quantifying adverbial in (38). This is particularly clear when *haufenweise* modifies an adjective (38c). As such, we could claim here that this element does not fall under the current generalization. These assumptions might explain this set of data – after all, adverbials cannot modify common nouns directly. However, it is unclear now why the (a)-type of examples in (37) above, where the determiner is absent, are grammatical: Why is an adverbial element modifying a common noun allowed here? As shown in Section 2, the quantifier and the noun do form a constituent. I will suggest that both (37) and (38) fall under the current generalization, with a justifiable qualification for (38b) above.

I assume with Pittner that the grammatical cases in the (a)-type of examples in (37) involve *haufenweise* as an adverbial element. In particular, I assume that *haufenweise* is left-adjoined to the noun phrase, perhaps similar to *nicht* ‘not’, *voll* ‘fully’, or *gestern* ‘yesterday’ below ((39c) is a googled example):

- (39) a. Nicht die Autos wollte ich sehen (sondern....)
 not the cars wanted I see but
- b. Voll der Idiot war er nicht.
 fully the idiot was he not
- c. Gestern das Treffen war wieder ein Highlight.
 yesterday the meeting was again a highlight

When *haufenweise* is inside the DP, as in the (b)-type of examples in (37), it is also adjoined and also falls under the current generalization. Given the indeclinable nature of adverbials, this explains the ungrammatical examples in (37b) and (38a).²¹ Now, in order to account for the grammaticality of (38b) above, I assume that adverbials can adjoin inside a noun phrase, namely, when a verbal projection is present as with nominalized verbs (for recent discussion, see Iordăchioaia 2020). Furthermore, observe that nominalized verbs are non-agreeing. Note in this regard that they cannot undergo plural formation (e.g., **die Horten(s)* ‘the hoardings’). This means that the agreement chain is not violated, as the last element of the chain, the noun, is –AGR. In other words, the chain is +AGR –AGR –AGR, and there is neither a disruption here nor are the two non-agreeing elements on the left.

6. Conclusion

The agreement chain explains why indeclinable quantifiers cannot be preceded by determiners. Furthermore, it explains why optional inflections on quantifiers in initial position become

²¹ It is possible that this type of example is independently out, given that such adjunctions of adverbs are usually not possible inside the DP. Be that as it may, the example is also ungrammatical due to the explanation offered below.

obligatory when a determiner is present. Given that, we can suggest that indeclinable quantifiers can be assumed to be located in Spec,CardP.

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